

Abdul Hakam

Electronic and Telecommunication Engineering Undergraduate

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Profile

Electronic and Telecommunication Engineering undergraduate focused on embedded systems, analog and digital electronics, PCB development, communications, and electronic product design, with interests in machine learning and computer vision. Experienced in developing and testing hardware-software systems using microcontrollers, sensors, custom PCBs, firmware, control logic, wireless communication, and system-level integration.

Education

B.Sc. Engineering (Hons.) in Electronic and Telecommunication Engineering **2024–Present**
University of Moratuwa, Sri Lanka *CGPA: 3.47/4.00 after four semesters*

GCE A/L Examination: Physical Science **2022 (2023)**
St. Anthony's College, Kandy *3A with a Z-score of 2.1072*

GCE O/L Examination **2019**
St. Anthony's College, Kandy *Results: 9A*

Experience

Owner and Operations Manager **2023–Present**
Gatronix Store 📍 Sri Lanka

- Own and manage Gatronix Store, a retail and wholesale computer-products business.
- Oversee purchasing, inventory, supplier and customer relationships, accounts, and day-to-day operations; also designed, developed, and now operate dedicated retail and wholesale business systems.

Technical Skills and Tools

- **Electronics and PCB Development:** Analog and digital circuit design, schematic capture, PCB layout, prototyping, soldering, and hardware debugging.
- **Embedded Systems and Interfaces:** STM32, ESP32, Arduino, Raspberry Pi.
- **Programming Languages:** C, C++, Python, Java, and JavaScript.
- **Simulation and CAD Tools:** Altium Designer, Proteus, LTspice, and SolidWorks.
- **AI, Machine Learning, and Computer Vision:** MediaPipe, OpenCV, and scikit-learn.
- **Software Development:** React, Spring Boot, Supabase, Flutter, HTML, and CSS.
- **Development, Testing, and Networking:** STM32CubeIDE, Arduino IDE, Git, Postman, Wireshark, and Cisco Packet Tracer.
- **Spoken Languages:** English, Tamil, and Sinhala.

Projects

Embedded Electronics and Product Design Projects

WaveSync - RF-Linked Alarm Clock 📡 **2026**
Digital logic, R-2R DAC, comparators, 433 MHz OOK, Altium Designer

- Co-developed a microcontroller-free alarm system that compares digitally encoded clock data with a user-set alarm and broadcasts the resulting trigger to synchronized remote receivers.
- Designed the 433 MHz OOK transmitter and receiver circuits and contributed to schematic development, PCB layout, hardware integration, and prototype testing.

Analog Automatic Gain Controller 📡 **2026**
NE5532, JFET voltage-controlled resistor, Proteus, analog signal processing

- Designed and simulated an analog audio automatic-gain-control circuit using op-amp buffering and amplification, peak detection, feedback control, and a JFET voltage-controlled resistor.
- Built and tested the hardware prototype with function-generator and real-audio inputs, improving output-level stability while reducing loading, clipping, and distortion.

Servie - Autonomous Restaurant Food-Delivery Robot  **2026**
ESP32-S3, STM32, C/C++, custom electronics

- Designed and integrated an autonomous restaurant delivery robot using magnetic-line navigation, encoder feedback, ultrasonic obstacle detection, tray load sensing, and motor-control electronics.
- Developed and tested the ESP32-S3 and STM32 control integration, including inter-microcontroller communication, navigation logic, battery management, and system-level hardware debugging.

FallTrack Smart Fall-Detection Watch  **2025**
ESP8266, MPU6050, MAX30102, Flutter, custom PCB

- Co-developed a wearable safety device that detects falls using motion and orientation data, monitors heart rate, and provides real-time feedback through an OLED display and local alerts.
- Integrated embedded firmware, Wi-Fi communication, a Flutter companion application, custom PCB development, battery-powered operation, and a compact wearable enclosure.

Software and Intelligent Systems Projects

Real-Time ASL Fingerspelling Recognizer  **2026**
Python, MediaPipe, OpenCV, scikit-learn

- Developed a real-time webcam system that collects normalized 3D hand landmarks and recognizes static ASL fingerspelling using a calibrated SVM, confidence filtering, temporal voting, rejection-pose training, and CPU inference optimizations.

Gatronix Retail and Wholesale Business Systems: Retail POS  | **Wholesale POS**  **2026–Present**
React, Vite, Supabase, JavaScript

- Developed separate systems for Gatronix Store's retail and wholesale operations, with workflows tailored to sales, purchasing, inventory, payments, and customer and supplier account management.
- Built retail workflows for POS billing, stock, quotations, returns, COD and online orders, cash flow, and documents; added wholesale workflows for bulk sales, supplier orders, goods receiving, stock in transit, landed cost, credit, cheques, and reporting.

University Lost-and-Found Management System  **2026**
React, Spring Boot, JWT

- Contributed to a full-stack campus platform for reporting, tracking, and recovering lost and found items through a centralized user and administrative system.
- Worked on frontend routing, authentication and user-related screens, backend integration, software testing, and supporting system documentation within the project team.

Competitions and Achievements

Robotics Design and Competition Module **2025**
Designed and built the best-performing robot for an arena-based navigation and task-completion challenge.

Sri Lankan Robotics Challenge (SLRC) **2025**
Participated in the university category of the national robotics competition.

SPARK 25/26 **2025–26**
Reached the semifinal stage with a flood early-warning system.

SLIoT Challenge **2026**
Reached the semifinal stage with a flood early-warning system.

OCTWAVE 3.0 – Kaggle Challenge 02 **2026**
Participated in the Tom & Jerry image-classification challenge, advanced to Round 2, and placed 16th.

ROSCO **2026**
Participated in the robotics competition.

Certifications

- **Data Science in Python** – Coursera, 2021.
- **Full-Stack Web Development Bootcamp** – Udemy, 2021.

Referees

Dr. Upeka Premaratne
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